



Computational Skills in Solving Application Problems Involving Basic Differentiation Rules in Differential Calculus: An Explanatory Sequential Study

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Suggested Citation

Booc, N.B.B., Ringcunada, E.J.D., Justiniani, A.M.Q., Arevalo, J.M.N., Chao Nui, J.P., Mora, R.C., Semblante, A.P. & Subingsubing, E.T. (2024). Computational Skills in Solving Application Problems Involving Basic Differentiation Rules in Differential Calculus: An Explanatory Sequential Study. *European Journal of Theoretical and Applied Sciences*, 2(1), 367-374. DOI: [10.59324/ejtas.2024.2\(1\).31](https://doi.org/10.59324/ejtas.2024.2(1).31)

Abstract:

This study aimed to identify the level of computational skills and the challenges of students in solving application problems using basic differentiation rules in differential calculus. This study employed a mixed method explanatory-sequential design, which involves collecting and analyzing quantitative data first, followed by the collection and analysis of qualitative data. In the quantitative phase of this research, a simple random sampling method was utilized to administer a modified questionnaire (problem-solving examination type) to 50 calculus students. In the qualitative phase, purposive sampling was used to administer semi-structured in-depth interviews (IDIs) to a sample of 6 participants. Mean and thematic analysis with document analysis were utilized to examine the information that helped researchers identify problem about the subject matter. The

study shows an overall high level of computational skills in basic differentiation, which means that the computational skills of students are often manifested. However, the computational skills of students in differential calculus in terms of chain rule are low, which is interpreted as rarely manifested. With this, this research had undergone an in-depth analysis of the challenges of the students in solving application problems using chain rule. The results reveal 3 challenges why the students' computational skills in terms of the chain rule are low: the complexity of the composition of the chain rule, a lack of practice and exposure in using the chain rule, and uncertainty regarding its application. Effective teaching strategies are essential for breaking down complex concepts and enhancing students' computational skills in basic differentiation rules in calculus.

Keywords: *Computational skills, basic differentiation rules, challenges in chain rule, differential calculus, college students, Davao City, Philippines.*

Introduction

Today's learners are striving to develop the necessary competencies for learning mathematics, with computational skills playing a crucial role in solving mathematical problems

(Ashraf et al., 2020). However, the complexity of teaching certain math concepts often leads to a lack of focus among students, hindering their ability to grasp necessary problem-solving steps and impacting their independent problem-solving skills. Computational fluency becomes



imperative, influencing retention in STEM-related undergraduate programs (Thompson & Harel, 2021). In various countries, deficiencies in understanding Differential Calculus concepts have been observed, with students struggling due to misconceptions and difficulties in connecting the fragments of pre-calculus and calculus concepts (Rosli et al., 2013; Tan & Shahrill, 2015). The Philippines, amidst the transition to the K to 12 curriculum, has experienced a significant drop in the achievement rate of first-year engineering students in Differential Calculus, raising concerns about the connection between pre-calculus and calculus concepts (Reyes, 2020; Gopalakrishnan et al., 2013; Bowman, 2019).

In Digos City, Davao del Sur, Philippines, low mean percentage scores in the 2014 National Career Assessment Examination highlight a broader issue of insufficient computational skills among students, impacting their ability to study higher mathematics effectively (Panerio, 2016; Gurat, 2018). The failure to instill the usefulness of mathematics in daily life further compounds the challenges in teaching mathematical abilities (Callaman & Auxtero, 2020). Addressing these issues requires a concerted effort to enhance instructional methods, bridge gaps in understanding, and emphasize the practical applications of mathematical concepts, ultimately empowering students to succeed in mathematics-related careers.

The purpose of this study is to determine the respondents' level of computational skills in solving application problems and experiences in basic differentiation rules of differential calculus. To be more specific, this study aims to answer the following

1. What is the level of computational skills on basic differentiation rules of students?
2. What are the challenges encountered by students in solving application problems using basic differentiation rules of differential calculus?

Methods

This research study utilized an explanatory sequential design. This research design is used to collect and analyze quantitative data, followed by the collection and analysis of qualitative data. The study was conducted in Davao City, Philippines, which offered Baccalaureate Degrees in engineering and mathematics education programs. In the quantitative phase, for the academic year 2022–2023, the researchers used the simple random sampling to select fifty (50) respondents who are currently enrolled in Calculus courses under the programs of Bachelor of Secondary Education major in Mathematics, Bachelor of Science in Computer Engineering, and Bachelor of Science in Electronics Engineering. In the qualitative phase, the researchers used the purposive sampling technique. Specifically, a sample of six (6) participants, consisting of two (2) from BSED Math, two (2) from Electronics Engineering, and two (2) from Computer Engineering programs, was selected to participate in the semi-structured In-depth interview (IDI). The participants were chosen based on having the lowest scores in the modified questionnaire (examination type), and the selection was done through a face-to-face setup. The value of ethical considerations in research was emphasized. Ethical research standards served as principles that protected morals and guided the researchers throughout the study.

Results and Discussions

Level of Computational Skills in Solving Application Problems of the Respondents

The table shows the level of computational skills of the respondents in terms of constant, power, sum and difference, product, quotient, and chain rule. It is also shown in the table the mean and descriptive level.

Table 1. Level of Computational Skills in Solving Application Problems Using Basic Differentiation Rules of Differential Calculus

Indicators	Mean	Descriptive Level
Constant Rule	3.74	Very High
Power Rule	3.28	Very High
Sum and Difference Rule	2.96	High
Product Rule	2.78	High
Quotient Rule	2.32	High
Chain Rule	1.18	Low
Overall	2.71	High

As presented in the table above, the overall mean under the level of computational skills of respondents is 2.71 with a descriptive level of high, which means that the computational skills of students in solving application problems using basic differentiation rules in differential calculus is often manifested. This implies that the observation of high computational skills in solving application problems using basic differentiation rules is a positive indication of students' mathematical proficiency. This contradicts to the idea of Larsen et al. (2017), which stated that students do not understand the quantitative meaning of differentiation and its real life application. Additionally, students do not have proficiency when it comes to differentiation due to their background and experiences in the prerequisite subjects (Maglipong et al., 2015). However, students' difficulties in differential calculus was considered by the National Council of Teachers of Mathematics (NCTM) as productive struggle in mathematics (DuCloux et al., 2018).

In addition, the mean of the *Constant Rule* under the level of computational skills of the respondents is 3.74 with a descriptive level of very high, which means that the computational skills of students in solving application problems using basic differentiation rules in differential calculus is always manifested. This contradicts the idea of Vrabel (2014), which stated that students in constant rule in differential calculus have difficulties in applying this rule, leading to a misperception that constant expressions like $y=5$ are not functions because there is no x .

However, these inconsistencies can be avoided by the learners if they have a shared understanding of what a constant rule is and how to represent it conceptually (Maharaj & Wagh, 2014).

However, the mean of the *Chain Rule* under the level of computational skills of the respondents is 1.18 with a descriptive level of low, which means that the computational skills of students in solving application problems using basic differentiation rules in differential calculus is rarely manifested. This implies that weakness in the chain rule may restrict students' ability to apply calculus principles in various fields and limit their potential career opportunities that require advanced mathematical skills. Also, lack of understanding in the chain rule could have long-term consequences, affecting students' success in higher-level mathematics and related disciplines. According to Roble (2017), students were not really caring about giving their differentiation any context because for them, it was as simple as following rules and not being aware of a certain function's derivative. This idea is similar to the study of Booc et al. (2023), who stated that students have low conceptual understanding, limited comprehension, and mastery of procedural fluency, emphasizing the need for improved mathematical skills. As a result, it was revealed the considerable conceptual gaps between the students' computational skills to carry out procedures based on those concepts and procedural knowledge of the primary ideas of calculus specifically chain rule (Mac an Bhaird et al., 2017).

Challenges of Students in Solving Application Problems Using Chain Rule Basic Differentiation Rules of Differential Calculus

The quantitative data reveals a pronounced deficiency in respondents' computational skills, particularly in the Chain Rule, as indicated by the challenges illustrated in the diagram based on participants' responses.

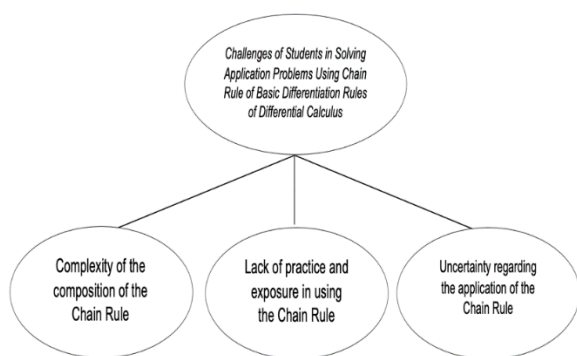


Figure 1. Challenges of Students in Solving Application Problems Using Chain Rule of Basic Differentiation Rules of Differential Calculus

First challenge is *the complexity of the chain rule*, which can lead to confusion and challenges for students. This means that students have difficulty in solving application problems, as understanding how changes in one function affect the overall function requires careful analysis and can be challenging to grasp. These are the evident answers of the participant:

“Na confuse ko unsaon pag identify sa given rate ug unsaon ba pag differentiate aning given sa chain rule gamit aning equation” (I get confused on the process of identifying the given rate and determining how to differentiate the given problem using the chain rule with this equation).

“Maglibog sad ko kung kanus-a ko mugamit ug chain rule kanang function of a function or product rule na function i-times sa function.” (I get confused when to use a chain rule, that is a function of a function or a product rule that is a function multiply to a function).

“Nag lisod ko ug recognize ug identify sa inner ug outer function” (I'm having a hard time recognizing and identifying the inner and outer functions.)

These challenges provided by the participants' statements are associated with the complexity of the chain. One participant finds it confusing to identify the given rate and determine how to

apply the chain rule to differentiate the equation appropriately. Another participant struggles with using the chain rule for functions of functions or when dealing with products of functions. Moreover, a different participant faces difficulty in recognizing and identifying the inner and outer functions involved in applying the chain rule. These challenges can be addressed through practice, a thorough understanding of the concept, and seeking clarification when needed.

According to the research, several previous research works have concentrated on investigating students' understanding of the derivative concept and methods to enhance their comprehension (Nobre et al., 2016). The findings from some of these studies indicated that many students encounter difficulty in grasping the concept of derivatives, especially when it comes to applying derivative rules to composite functions. The analysis of the question in this study revealed a facility index of 82.54%, suggesting that the question was generally considered easy by the students. The subsequent question in the study focused on the composite function $y = [g(x)]^n$ and aimed to understand the specific challenges faced by students finding the derivative of composite functions (Maharaj, 2013). In the same study, revealed that they had difficulty in differentiating the function $h(x)$. A possible reason for this difficulty could be that the function $h(x)$ has a structure which requires the application of the product and chain rules for differentiation, and more than one application of the chain rule (Park & Lee, 2016).

Second challenge is the lack of practice and exposure in using the chain rule, which can lead to students struggling to apply it effectively in problem-solving. This means that without sufficient experience and familiarity with the chain rule, students may find it challenging to understand its concepts and confidently utilize it in various application problem. These are the evident answers from the participants:

“Wala kay koy knowledge about sa constant rate of change sa chain rule” (I don't have any knowledge about the constant rate of change in the chain rule).

"Wala kaayo koy knowledge about sa chain rule ug wala kayo nako na practisan" (I don't have enough knowledge about the chain rule and haven't had enough practice with it).

"Wala pa sad koy enough knowledge about functions og constant rate of change sa chain rule" (I do not have yet enough knowledge about functions and the constant rate of change of chain rule).

Based on the participants' responses, the challenges related to the lack of practice and exposure in using the chain rule are evident. One participant admits to not having enough practice and exposure with the chain rule, which suggests a potential lack of confidence in applying it to various mathematical problems. Additionally, other participants mention a deficiency in their understanding of the chain rule itself, indicating a limited grasp of its principles and applications. Moreover, another participant highlights their insufficient knowledge about functions and the constant rate of change within the chain rule context. These challenges collectively emphasize the need for more practice, exposure to diverse problems, and a deeper understanding of relevant concepts to overcome these obstacles effectively.

It seems that if students could first practice and engage to the given structure of the function, then detecting and noting the structural representation could help them to decide how to apply basic differentiation rules when finding the derivative of the relevant function (Maharaj, 2013). However, the traditional approach which puts emphasis on computational procedures rather than on understanding the underlying concepts is still the most commonly used method by teachers (Lasut, 2015). As a result, many students fail to manifest excellent performance because they lack practice and exposure to chain rule and its concepts in real life situation (Fluck & Dowden, 2013).

Third challenge is the uncertainty regarding the application of the Chain Rule, which can lead to confusion and difficulties for students. This means that students may face challenges in effectively utilizing the Chain Rule to solve problems, as they lack a clear understanding of

when and how to apply it. These are the evident answers provided by the participants:

"Nalibog ko kung tama ba jud nga $A=r^2$ iyang equation, kung ang r^2 ba niya is 1.56 or 5.4 feet" (I doubt if the equation $A = r^2$ is correct, and if so, whether ' r^2 ' should be 1.56 or 5.4 feet).

"Dili ko kabalo mo solve ana gamit ang chain rule kay wala pa kaayo nako na sweto." (I don't know how to solve using the chain rule because I'm not familiar with it).

"Maka confuse jud na part is ang equation na $A=r^2$ kung tama ba sya ang gamiton nga equation" (Confusing part is that the equation $A=r^2$, if it is the correct equation to solve).

The challenges arising from the uncertainty regarding the use of chain rule are evident in the participants' responses. One participant expresses doubt about the correctness of the equation $A = r^2$ and is uncertain whether ' r^2 ' should represent 1.56 or 5.4 feet. Another participant admits their inability to apply the chain rule due to unfamiliarity with it. The confusion surrounding the equation $A = r^2$ adds to the difficulties faced in attempting to solve problems using the chain rule. These responses underscore the importance of practicing and gaining exposure to mathematical concepts like the chain rule to enhance problem-solving skills and build confidence in tackling related equations and calculations effectively.

According to this study students' uncertainty surrounding the application of the chain rule in the context of teaching and learning the derivatives of exponential functions. The student's incorrect response to a particular question indicated that they could recognize the need for the chain rule in certain situations but not in others (Park et al., 2016). The findings suggest that successful differentiation goes beyond merely applying rules and algorithms; it involves a deeper understanding of derivative-related concepts and the reasons behind the steps in the algorithm (Fluck et al., 2013). To enhance the teaching and learning of derivatives, there should be a focus on understanding the

underlying concepts and interactions based on the symbolic representation of functions or equations. This includes exploring the structure of composite functions and identifying when the chain rule is required, which proved to be a discriminating factor in successful answers (Nobre et al., 2016).

These are also evident to the answers of the participants in the problem solving examination particularly item number 6. One of the participants, was only able to write the given values including the formula $A = r^2$, but was not able to identify the inner and outer functions or perform any computation that involves chain rule. Also, the other participant just wrote down without even labeling what those values represent. While the other participant left the answer key blank. Moreover, one of the participants was not able to differentiate the formula of $A = r^2$. However, the other participant was able to write only $A = [(1.56)(5.4)]^2$ but did not differentiate the formula $A = r^2$ which leads to wrong answer. And other participants were not able to multiply first before substituting the values 1.56 and 5.4 feet to get the value of r and forgot or did not able to differentiate the formula $A = r^2$.

This idea is supported by the study Weber (2013), who claims that college students often struggle to apply the guidelines of the chain rule effectively, resulting in computational errors. It is also supported by Jojo et al. (2013) and reveals that South African students faced difficulties in accurately performing computations due to a lack of understanding of the "o" symbol representing function composition. Additionally, Roble (2013) supports this idea by highlighting the difficulties some students encountered in correctly differentiating trigonometric functions, specifically cosine ($\cos x$). Furthermore, Jaafar and Lin (2017) provides further support this idea by finding that students made the error of incorrectly stating that the derivative of $\cos(x^2)$ is " $2 \sin(x) \cos(x)$ " due to their inability to identify the outer and inner functions. Moreover, Wahyuni et al. (2017) emphasizes that students frequently face challenges when learning the basic concepts,

principles, and skills of differential calculus. This leads to incorrect answers or incomplete solutions, as some students perceive the problem as more complex in chain rule, causing them to get stuck and abandon their attempts.

Conclusion

The research findings indicate a high level of computational skills among students in basic differentiation, particularly in the constant rule, power rule, sum and difference rule, product rule, and quotient rule. However, there is a noticeable deficiency in computational skills related to the chain rule. The mean and standard deviation suggest that students rarely manifest proficiency in applying the chain rule, with the qualitative phase identifying three primary reasons: the complexity of the rule's composition, a lack of practice and exposure, and uncertainty regarding its application. These results underscore the necessity for instructional approaches that simplify the chain rule, breaking it down into manageable components to enhance students' comprehension and application. Strategies such as increased practice opportunities, clear explanations, and targeted demonstrations are crucial to addressing the identified challenges and improving students' overall computational skills in basic differentiation within the realm of mathematics education.

Recommendations

The study suggests that the Commission on Higher Education should implement guidelines supporting the integration of targeted instructional strategies and resources into higher mathematics education curricula, employing a progressive learning approach for the chain rule. This involves starting with simple compositions and gradually progressing to more complex ones to build a solid foundation. The adoption of Problem-Based Learning in meaningful contexts and Interactive Learning Tools, like online simulations, can enhance students' understanding and application of the chain rule. School administrators should focus on faculty

development through monitoring, classroom observations, and constructive feedback. They can also promote effective teaching practices by relating the chain rule to real-life scenarios, offering support to students, and organizing workshops. Math instructors should simplify the chain rule's complexity through clear explanations, demonstrations, and exercises while creating a supportive learning environment. Students, in turn, should seek clarification, engage in regular practice, explore additional resources, and collaborate with peers. Future researchers can contribute by exploring alternative research designs and providing theoretical frameworks to address uncertainties in the application of the chain rule.

Conflict of Interests

No conflict of interest.

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